a

ECD structure + sequence

1. Residue mapping + preparation

Chain selection · WT check · normalized context + residue map

mapped context

2. Accessibility + single-site screening

ProCaliper

Native SASA comparison
SASA / rSASA cross-check

DiscoTope-3.0

Optional soft
epitope prior

SPURS single

Single-site
stability screen

single-site options

3. Sectors + mutation combination selection

Connected geodesic surface sectors

2/3-site candidate search

SPURS multi

Combination stability

FoldX

Combination confirmation

Stability + extrapolation gates

Deterministic panel selection

selected panel

4. Panel results

Selected panel + audit outputs
Formal / blocked / preview status